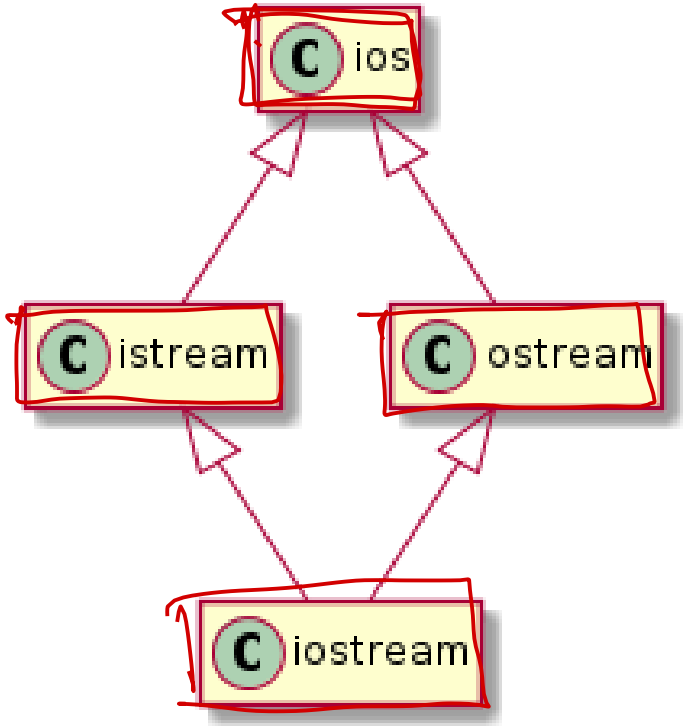
Sunbeam Infotech

Exploring new ideas, Reaching new heights!

C++ File I/O

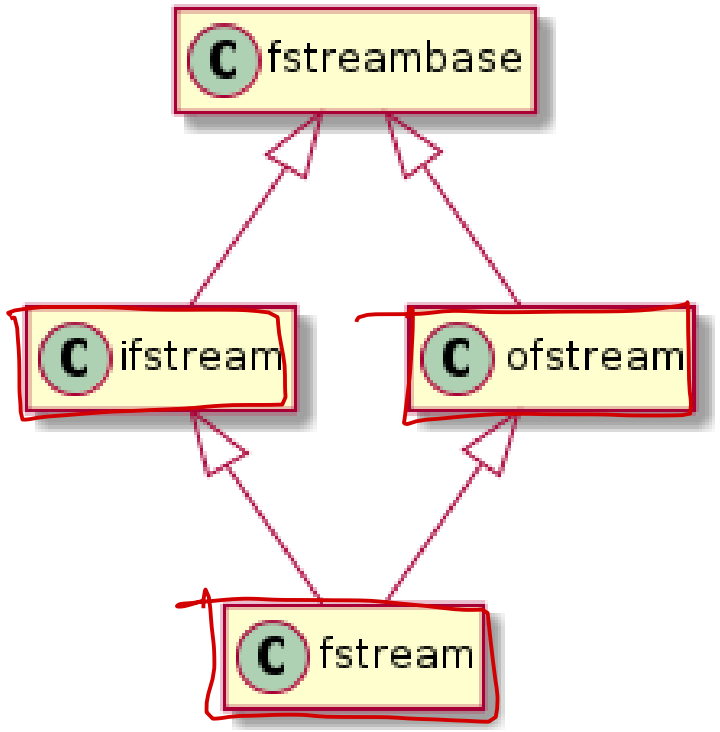
- Stream is a logical device that either produces or consumes information.
- stream is linked to physical device by I/O system (OS)
- All stream behaves in same way irrespective of device attached to it.
- Stream is continuous flow of bytes.
- These are standard C++ streams:
  - cin (istream) → stdin
  - cout (ostream) → stdout
  - cerr (ostream) → stderr
  - clog (ostream)

terminal (console)



C++ File I/O

- ifstream is used to read from file.
- ofstream is used to write into file.
- fstream is used to read & write into file.



- ifstream fin("filepath");
- ofstream fout("filepath");

- fstream file;
- file.open("filepath", mode);

- mode:
  - ios::in, ios::out, ios::app, ios::trunc, ios::binary

Binary file

short i=12345;

↓

2 bytes.

Text file

short i=12345;

↓

5 chars

(5 bytes).

C programming

* fread() & fwrite() → file position → fseek() ; tell()

C++ programs

* read() & write() → file pos.

tellg() / seekg()

ifstream → get pos.

ofstream → put pos.

tellp() / seekp()

C++ File I/O

- File IO
  - Formatted IO
    - operator<<() and operator>>()
    - Manipulators
      - boolalpha, showpos, hex
  - Char IO: get() and put()
  - String IO: getline()
  - Binary IO: read(), write()
  - Change file current position
    - seekg(), seekp()
    - tellg(), tellp()
  - Functions: open(), eof(), …

Thank you